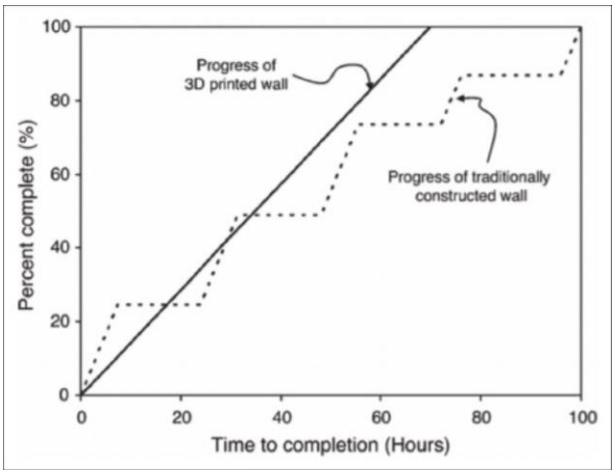
## Introduction

In a disaster scenario, many homes are destroyed or damaged beyond repair, leaving survivors without a shelter to protect them from the elements. Construction companies using traditional building techniques work as fast as they can but are unable to help the potentially thousands of people affected quickly enough. With recent advantages in the technology of concrete 3D printing, however construction speed has greatly increased. This leads many to ask the question, “In what ways are robot arm concrete 3D printers more efficient at constructing houses as opposed to traditional housing?”. In this formal report, the application of 3D Concrete Printing (or 3DCP for short) as a solution for disaster relief and reconstruction is focused upon. Traditional house constructions such as brick-and-mortar often fail to be deployed in time for emergency housing, and this is where 3DCP comes in and provides a viable alternative. This report will demonstrate the greater mobility and reduced deploy time of the robotic arm system, being the more efficient tool for rapid housing deployment in post-disaster environments.

## Methods

The method for testing our robotic arm will be to print multiple houses with a high precision rate in order to ensure high quality service. This will cover percent of errors of the prints, the speed of the printer, frequency of maintenance, and how much it can print in an allotted time. We will be doing secondary research using research reports such as “Multi-criteria decision framework for optimal robotic system selection in 3D concrete printing” [1] to validate our results.

## Results

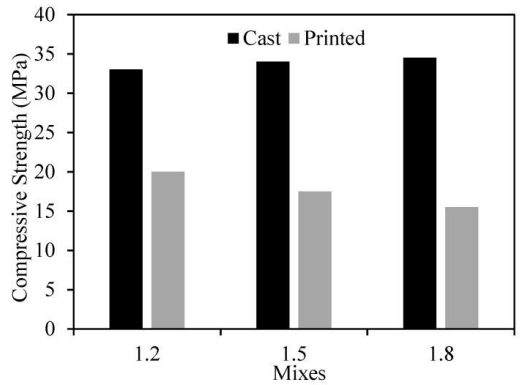



In accordance with rigorous compression, penetration, ductility, flow, and slump testing, many results in comparison to cast concrete have been obtained. In brief, 3D printed concrete has many downsides in comparison to conventional cast concrete, the main point being the loss of structural stability. These downsides are negated by a faster build speed, less manpower required, and the ability to use prefab design templates. Most 3D printed concrete mixtures dry in about 75 hours whereas it takes around 2 to 4 weeks for a traditionally built modern home.

While it is faster to construct houses, the concrete loses its compression resistance, which wouldn’t allow buildings to be much taller with minimal support. These issues can be mitigated by using additional additives, such as sand, silica fume, and blast furnace slag. Through testing, the optimal mix, 1.2, for 3D printing of concrete is a mix of ground granulated blast furnace slag, silica

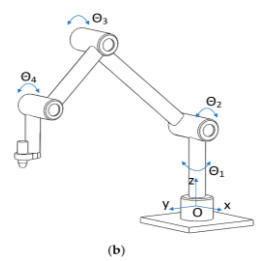
fume, dune and crushed sand, all combined with the control mix of standard concrete [2]. In using these additives, we can produce a faster print speed, allowing the construction of houses to be expedited. This brings the total build time, after initial installation of the concrete base, to about 2 (two) weeks.
With the addition of a square 20 x 20mm nozzle and the additives previously mentioned, testing results are very close to conventional cast concrete.

However, VMA (viscosity modifier admixture) is required for the mixture to be extruded [2]. VMA increases the cost of the concrete mixture, but increases its performance in printing, allowing buildings to be constructed faster, at a slightly lower compression resistance rating than cast concrete. Cast concrete after 3 days of setting, results in a compression resistance of 34 MPa, and 3D printed concrete at the same time only results in 20 – 15.5 MPa [2], measured in Megapascals. The resulting performance indicates the areas that require improvement in our mixtures.




## Discussion

The range of motion of the robotic system is more ideal for a small-scale production project. It is quicker to move from site to site than the alternatives, making for ease of access in a disaster scenario. As seen in figure 1, the stationary arm is limited to its vicinity and is limited by height (ranges will vary from printer to printer). The mobile arm, while still having these limitations, can be easily moved to improve the range of access.




Using concrete mix 1.2[2], we can maximize compression resistance. While the compression resistance of 3DP concrete mixtures is lacking in comparison to casted concrete it is sturdy enough to withstand the elements and provide shelter. This mixture also dries much faster compared to cast concrete, finishing in just roughly 75 hours or 3 days, making the speed of production quicker overall. However, one of the limitations we encountered during our research was that due to the technology being newly emergent, there were fewer peer reviewed research papers surrounding it making for more scattered results. In conclusion, we have found it a more efficient form of construction that still compares to the quality of current standards and would prove invaluable in a disaster scenario.

## Works Cited

[1] C. Lapusan, C. M. Negrutiu, and I. Sosa, “Multi-Criteria Decision Framework for Optimal Robotic System Selection in 3D Concrete Printing,” Buildings, vol. 16, no. 1, p. 202, Jan. 2026, doi: https://doi.org/10.3390/buildings16010202.

[2] S. Ahmed and S. Yehia, “Evaluation of Workability and Structuration Rate of Locally Developed 3D Printing Concrete Using Conventional Methods,” *Materials*, vol. 15, no. 3, p. 1243, Feb. 2022, doi: https://doi.org/10.3390/ma15031243.